

(1) Suppose $f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is a linear transformation and

$$f(1, 0, 0) = (1, -3), \quad f(0, 1, 0) = (-2, 6), \quad f(0, 0, 1) = (0, 0)$$

(a) Find $f(3, 4, 5)$.

$$\begin{aligned} f(3, 4, 5) &= 3(1, -3) + 4(-2, 6) + 5(0, 0) \\ &= (3, -9) + (-8, 24) = \underline{(-5, 15)} \end{aligned}$$

(b) Find a matrix A such that $f(x) = Ax$, for all x in $\mathbb{R}^3$.

$$A = \begin{bmatrix} 1 & -2 & 0 \\ -3 & 6 & 0 \end{bmatrix}; \quad \text{Check: } \begin{bmatrix} 1 & -2 & 0 \\ -3 & 6 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 - 2x_2 + 0x_3 \\ -3x_1 + 6x_2 + 0x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ -3 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \end{bmatrix} \checkmark$$

2×3

(c) Find the kernel and range of f

$$\text{range } f = \text{span} \{ (1, -3) \} = \text{column space of } A.$$

$$\text{kernel } f = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\} = \text{Null space of } A$$

(2) Suppose the function $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined as $g(x_1, x_2) = (2x_1 + 3x_2, x_1 + 4x_2)$.

(a) Compute $g(3, 1)$.

$$g(3, 1) = (2 \cdot 3 + 3 \cdot 1, 3 + 4 \cdot 1) = (9, 7)$$

(b) Show that g is a linear transformation by finding a matrix A such that $f(x) = Ax$, for all x in $\mathbb{R}^2$.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}; \quad \text{So } \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + 3x_2 \\ x_1 + 4x_2 \end{bmatrix} \checkmark$$

$$g(x_1, x_2) = (2x_1 + 3x_2, x_1 + 4x_2).$$

- (c) Compute $g(3, -1)$ and $g(1, 1)$. What does this tell you about matrix A ? (List as many answers to this question as you can think up!)

$$g(3, -1) = (2 \cdot 3 + 3(-1), 3 + 4(-1)) = (3, -1)$$

$$g(1, 1) = (2 + 3, 1 + 4) = (5, 5)$$

A has eigenvalues 1 and 5

A has eigenvectors $(3, -1)$, $(1, 1)$.

A is diagonalizable, and invertible.

$$N(A) = \{0\}$$

$$C(A) = \mathbb{R}^2$$

- (d) Find a (different) matrix, say B , such that $g(\mathbf{x}) = B\mathbf{x}$ under the assumption that all vectors in $\mathbb{R}^2$ are written in terms of the basis: $\mathbf{v}_1 = (3, -1)$, $\mathbf{v}_2 = (1, 1)$.

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}. \quad \text{This is because all other vectors } \vec{w}$$

$$\text{would be written as } \vec{w} = c_1 \vec{v}_1 + c_2 \vec{v}_2 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$\text{So } g(\vec{w}) = B\vec{w} = B(c_1 \vec{v}_1 + c_2 \vec{v}_2) = c_1 \cdot \vec{v}_1 + c_2 \cdot 5 \vec{v}_2. \text{ or}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ 5c_2 \end{bmatrix} \quad \checkmark$$

- (e) Use the matrix B to find the image of $\mathbf{w} = (3, 1)$ under g and confirm it is the same as your computation in part (a).

• We need to write $\vec{w}$ in terms of $\vec{v}_1$ and $\vec{v}_2$. So

$$c_1(3, -1) + d(1, 1) = (3, 1) \quad \vec{w} = \frac{1}{2} \vec{v}_1 + \frac{3}{2} \vec{v}_2$$

$$3c + d = 3$$

$$-(-c + d = 1)$$

$$4c = 2$$

$$c = \frac{1}{2}, \quad d = \frac{3}{2}$$

$$g(\vec{w}) = B \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{15}{2} \end{bmatrix} = \frac{1}{2} \vec{v}_1 + \frac{15}{2} \vec{v}_2$$

$$= \frac{1}{2}(3, -1) + \frac{15}{2}(1, 1) = (9, 7) \quad \checkmark \quad \text{!!}$$